Analysis of Virginia Traffic Crash Patterns & Economic Correlations

BUAD 5272: Database Management

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Contents

1 Introduction

1.1 Business Problem

Traditional crash analysis often treats accidents as isolated incidents, focusing solely on individual driver behavior or vehicle mechanics. However, this approach overlooks the broader socioeconomic context that shapes road safety.

Our project aims to bridge this gap by integrating crash data with economic indicators in each Virginia county, specifically examining individual poverty levels. By shifting the focus from particular cases to regional trends, we can assess whether specific economic zones face disproportionate risks due to area road conditions—such as poor lighting and roadway defects. This analysis allows for more targeted and equitable infrastructure improvements.

1.2 Data Sources

We integrated two primary datasets for this Data Warehouse:

1. **Virginia Poverty Data:** Sourced from the HDPulse Data Portal (NIMHD). This dataset tracks the percentage of persons living below 150% of the Federal Poverty Level across Virginia counties.

2. **Virginia Crash Data:** Sourced from the Virginia Department of Motor Vehicles (DMV) Open Data Portal. It contains granular details of traffic accidents, including severity, driver behavior, and environmental conditions.

2 Dimensional Modeling

2.1 Initial ERD (Draft)

In our initial conceptualization, we designed a schema where `DIM_Location` served as the central hub (acting as a pseudo-Fact table). We attempted to link disparate transactional data, such as `Crash_ID` and `Income_ID`,directly to specific jurisdictions.

However, this design presented a critical **granularity mismatch**. Locations are static reference entities (low granularity), whereas traffic accidents are high-volume, dynamic events (high granularity) occurring at specific timestamps. By centering the model on Location, we inadvertently created a "many-to-one" trap that made time-series analysis impossible; we could not easily query crash trends over months or years because the schema was optimized for describing *places*, not tracking *events*. This realization forced us to pivot to a transaction-grained Fact table (`FACT_CRASH`) to correctly capture the temporal dimension of road safety.

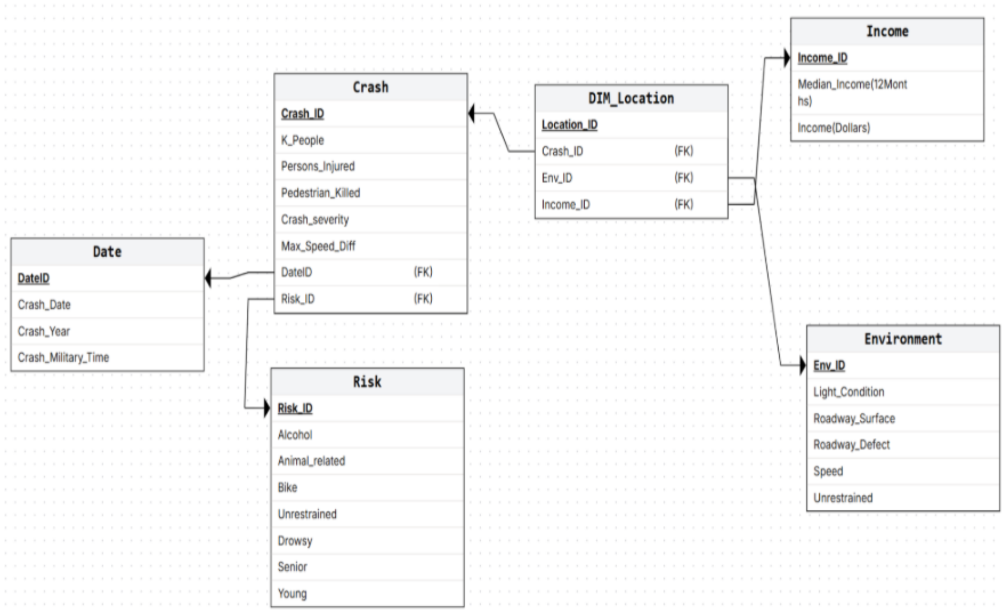


Figure 1: Initial Entity Relationship Diagram.

2.2 Final ERD (Implementation)

We pivoted to a robust **Star Schema** architecture centered on a transactional fact table. In this design, the grain of the fact table represents a single crash event, preserving the highest level of detail. We also added a "rollup" fact table called fact locality summary. Within the time dimension, we incorporated rows to represent the whole year (e.g., 2017, NULL, NULL, "full year") to enable linking the fact locality summary table to the time dimension. This is called a multi-grain dimension and is ideal for our needs.

Crucially, we integrated the socioeconomic data into an enriched DIM Locality table. By denormalizing poverty rates and economic tiers into this dimension, we enabled seamless "slicing and dicing" of crash metrics against community wealth levels. This structure also includes specialized dimensions for Severity, Time, and Roadway Defects, allowing us to isolate specific environmental risk factors (e.g., unlit roads) without duplicating data across millions of rows.

3 ETL Processes

3.1 Workflow Description

We utilized **Alteryx** to perform the Extract, Transform, and Load (ETL) processes. The workflow involved the following key steps:

- **Data Cleaning:** We utilized Regex tools to standardize locality names (e.g., converting "Richmond City" formats) to ensure successful joins between the Crash and Poverty datasets.

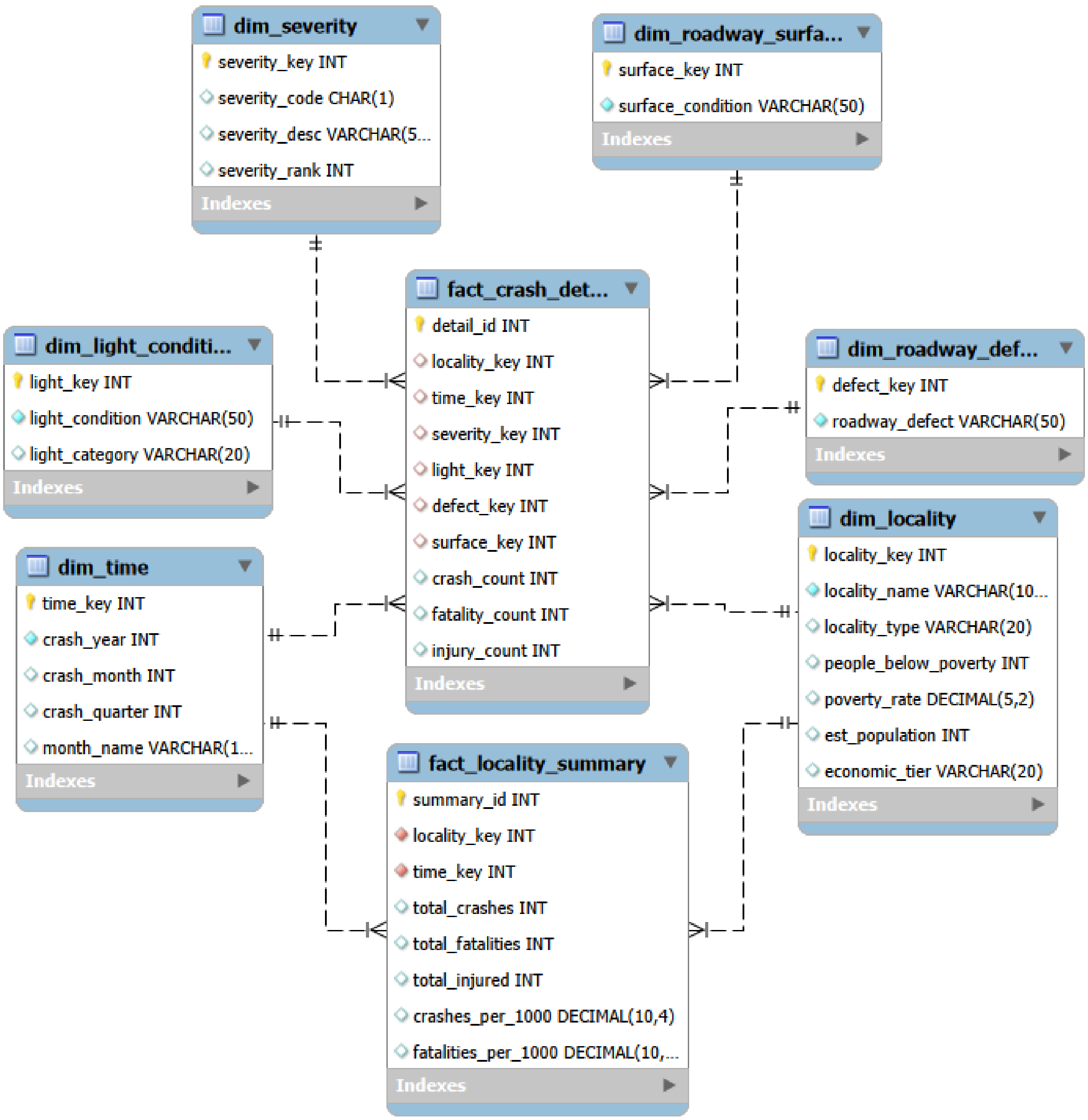
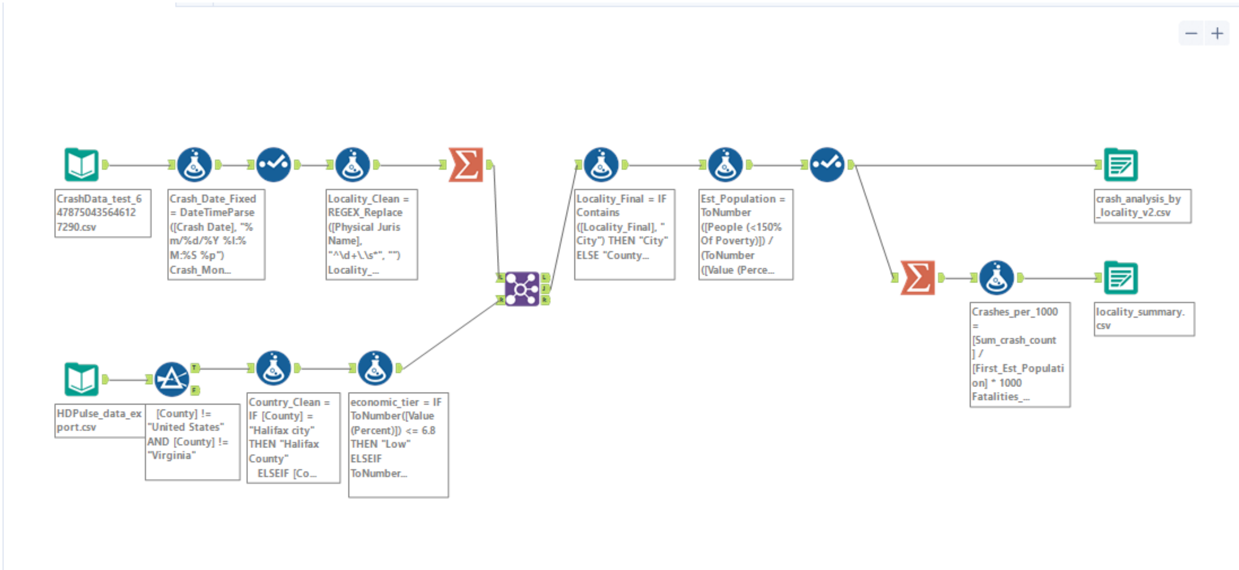


Figure 2: Enter Caption



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- **Metric Creation:** We calculated per-capita rates, creating new fields for "Crashes per 1,000 population" and "Fatalities per 1,000 population".
- **Economic Tiering:** We created a conditional logic formula to categorize localities into "Low", "Medium", and "High" tiers based on the percentage of the population below the poverty line.

4 Analytics & Findings

4.1 Crash Frequency by Economic Tier

Although we started the project with the intent of letting the data tell the story in an unbiased way, it might seem logical that areas with higher proportions of economically disadvantaged people would be lower priorities for public works departments. Oftentimes, wealthier areas have nicer roads and lighting because the people who live there are more adept at navigating the systems that provide those services. Therefore, places with better upkeep should have a lower accident rate due to safety-related factors, such as those mentioned. Surprisingly, the data did not show us what we expected. It revealed that higher-income localities (labeled "High" tier) experience more crashes per capita (approx. 128 per 1,000) than lower-income areas.

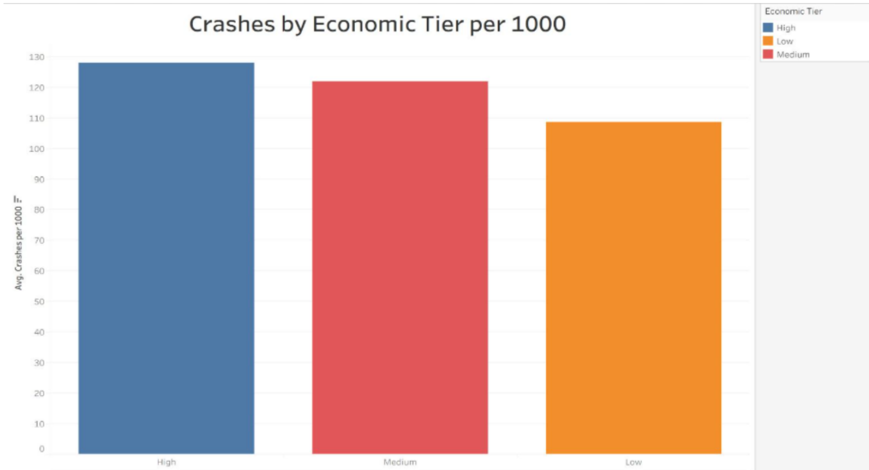

Figure 4: Average Crashes per 1,000 People by Economic Tier.

4.2 Impact of Poor Road Conditions

Shifting focus to road conditions, we filtered the data to include only road conditions identified at the crash site that are repairable by public works, such as potholes or significant road defects. This excludes many other things, such as ice or other weather-related artifacts. Once again, our analysis revealed that higher-income localities (labeled "High" tier) experience more crashes per capita than lower-income areas. However, the difference is statistically insignificant, with 1.2 per 1000 versus 1.1 per 1000.

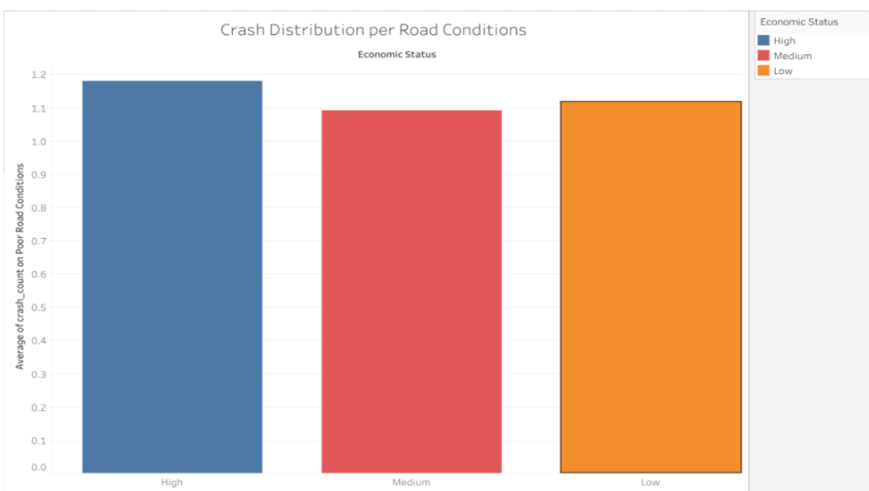

Figure 5: Average Crashes on Poor Road Condition

4.3 The Impact of Lighting in Low-Income Areas

Moving on to lighting impacts at crash sites, we filtered the accidents to show only those occurring at night, when lighting might have an effect. While the data show a continuing trend of fewer accidents in well-lit, poorer areas, we do see a reversal in that trend with more accidents in the poorer areas with no light at the accident scene. The number of accidents jumped from 3 per 1000 to 5.5 per 1000 in poorer areas with the roadway ”not lighted”. The not-lighted category could be due to lamps being out or no lighting ever being installed.

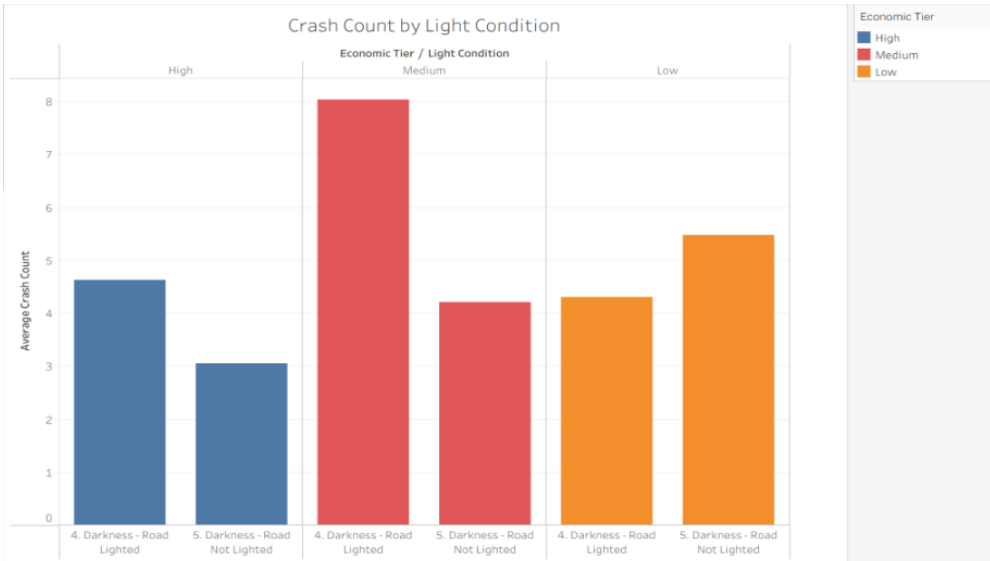

Figure 6: Crash Counts by Light Condition and Economic Tier.

5 Conclusion

5.1 Recommendations

Clearly, a goal of zero accidents is desirable but not achievable. Given differences in accident rates and the reversal in lighting-related trends, we recommend that Virginia counties prioritize roadway lighting improvements, particularly in economically distressed areas. The data indicate that such investments could have the most significant marginal safety impact in these communities.

5.2 Reflection

- **Challenges:** The most difficult step was cleaning the data sufficiently to implement it into the ERD, particularly matching locality names across different government sources.
- **Successes:** Developing the business idea was straightforward, as we utilized the Virginia Open Data portal to find relevant, non-political datasets.
- **Key Lesson:** We learned that data does not always tell the story you expect. Looking at sheer numbers, there were many, many more crashes in higher-income areas. We would guess this is simply because of the higher number of vehicle owners, but that analysis is for another study. This showed us the importance of data normalization, careful variable selection, and thoughtful filtering.